Chapter-8

(8.1) Fundamentals of thermodynamics

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Introduction:

When two stones strike together spark may be produced. ~~When two dry woods are rubbed together fire may be~~ When we rub our hands each other, we fee warm. In both cases heat is produced in the expense of mechanical ~~work~~ energy. Also, when certain mass of ice falls from a height, it may melt. This is due to conversion of gravitational potential energy into heat energy. We have other examples such as in cars, bikes, airplanes etc. fuel is burnt to produce heat which causes the expansion of gases inside the cylinder and hence mechanical work is obtained. In all cases, we mentioned the principle of conservation of energy is not violated. Transformation of heat into mechanical work and vice-versa has provided us the ~~found~~ foundation for the construction of heat engines (like bike, scooter, car etc) and refrigenerators. Thermodynamics is the branch of physics which deals with the interrelationship between heat and mechanical work. Now a days thermodynamics has been more applicable in other fields like electrical, chemical and magnetic fields.

Note:

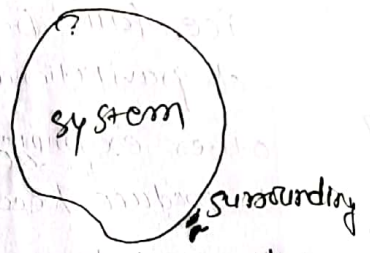
Thermodynamics ⇒ Thermo + dynamics = Heat + Study of motion.
It is the branch of physics which deals with the transformation of heat energy into mechanical work and vice-versa. Thermodynamics deals macroscopically i.e. it deals with the bulk system but not with the individual molecule.

Thermodynamic System (Types of thermodynamic system) & its

An assembly of large number of particles bounded by a closed surface (boundary surface) having certain value of

macroscopic parameters (pressure, volume and Temp) is called Thermodynamic system. Anything outside the boundary surface of system is called its surrounding. Thermodynamic system interacts with its surrounding either by exchanging energy and matter in different ways or by performing work.

On the basis of exchange of energy (heat) and matter, systems are classified into three types. They are:



[fig: Thermodynamic system]

① Open system: Thermodynamic system which can exchange both matter and energy with its surrounding is known as open system. eg. gas contained in an open cylinder.

② Closed system: Thermodynamic system which can exchange only energy but not matter with its surroundings is called closed system. eg. gas contained in a ~~perfectly~~ closed and conducting cylinder fitted with piston.

③ Isolated system: Thermodynamic system which can exchange neither energy nor matter with its surrounding is called Isolated system. eg. gas contained in a perfectly insulated cylinder fitted with piston.

Thermodynamical Variables and properties

An equilibrium state of a thermodynamic system is specified by the value of its parameters like pressure (p), Volume (V), and Temp. (T) etc. These parameters are called Thermodynamic Variables.

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On the other hand, thermodynamic property is any characteristics of thermodynamics variable and other additional terms like mass, internal energy, total energy, enthalpy, entropy, viscosity, thermal conductivity etc.

Thermodynamic properties are generally classified into two groups. They are:

① Intensive properties: The properties which are independent of the mass of the system are called intensive properties. eg. Temp. pressure, ~~density~~ etc. are examples.

② Extensive properties: The properties which depend on the size or extent of the system are called extensive properties. eg. Total mass, total volume and ~~total~~ momentum

II Thermodynamic equilibrium:

Thermodynamic deals with equilibrium states. Equilibrium means a state of balance. An equ^m state there is no existed unbalanced potentials within a system. A system will be in a state of thermal equ^m, if the condition for the following three types of equ^m are satisfied.

① Mechanical equ^m: A system is in mechanical equ^m if there is no change in pressure at any point of the system with time.

② Thermal equ^m: A system is said to be in thermal equ^m if all parts of it exhibit the same value of temp.

③ Chemical equ^m: A system is in chemical equ^m if its chemical composition does not change with time, there is no chemical reaction occur.

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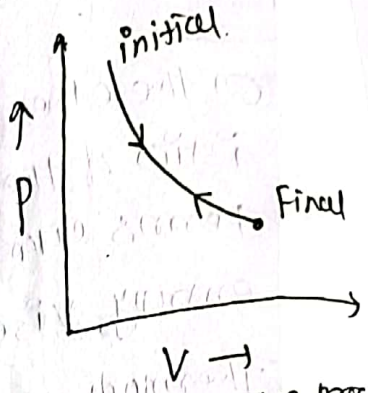
~~Thermodynamic process~~ (Reversible and Irreversible process).

(a) **Reversible process:**

If the effects produced by a process on the system as well as on the surrounding can be completely restored to their initial states and no changes are left in any of the system taking part in the process or in the surrounding then the process is said to be a reversible process. Any process, taking place slowly and controlled manner. All the mechanical process taking place under the action of conservative force are reversible process. In reversible process, a system can retrace in the opposite dir^n so that it passes through exactly the same states in all respect as in the direct process.



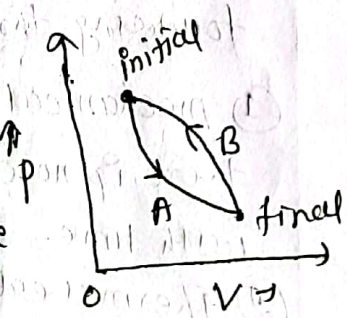
[Fig: Reversible process]

Conditions for the reversible process

1. The process should be infinitely slow
2. There should be no loss of energy during the process.
3. System and surrounding should be in thermodynamic equ^m during the process.
4. No dissipative force of friction, viscosity, electrical resistance etc should be presented.

(b) **Irreversible process**

The process which produce a permanent change in the thermodynamic state of the system and cannot be retraced in the opposite order are called irreversible process. ~~All most~~ Almost all processes are in the nature of irreversible process.



[Fig: irreversible process]

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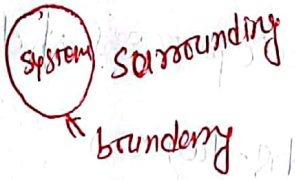
Eg. rusting of iron, decay of matter, heat produced by friction etc. are irreversible process.

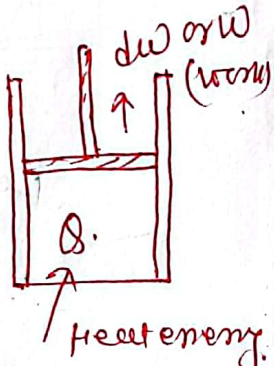
(8.2) Laws of thermodynamics

First law of thermodynamics is based on the principle of conservation. It states that "energy can neither be created nor can be destroyed but can change one form to another form". This law can be expressed in closed and open systems. For a closed system 1st law of thermodynamics can be explained based on the energy conservation. In open system, it can be explained based on the mass conservation and energy conservation. They are described as following:

First law of thermodynamics for closed system (For control mass):

In the closed system, energy can be transferred between system and surrounding but not matter.





Consider a closed system, amount of heat energy Q is given as an input and external work done (W) as an output as shown in figure.

Using the law of conservation of energy.

Energy entering the system − Energy leaving the system = Change in total energy of the system.

i.e. $E_{in} - E_{out} = \Delta E$

$$Q - W = \Delta E \quad \text{—(1)}$$

The total energy of the system (i.e. change of energy is the sum of K.E., P.E and internal energy.)

$\therefore \Delta E = \Delta K.E + \Delta P.E + \Delta U$ —(2) where ΔU = Change in internal energy of the system.

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In the case of closed system, change in K.E. and P.E are negligible as compared to change in internal energy. Therefore, from eqn (2), we get.

$$\Delta E = \Delta U \quad \text{— (3)}$$

Using eqn (3) in eqn (1) we get

$$Q - W = \Delta U$$

$$\text{or } dQ - dW = dU$$

$$\text{or } dQ = dU + dW \quad \text{— (4)}$$

This is an eqn. of 1st law of thermodynamics.

Case-1: If the closed system undergoes in cyclic process:

The conservation of energy eqn can be written as,

$$\oint (dQ - dW) = \oint dU$$

For cyclic process $\oint dU = 0$

$$\therefore \oint (dQ - dW) = 0$$

$$\text{or } \oint dQ = \oint dW$$

Therefore, net amount of applied heat energy (Q) is equal to net amount of external work done (W).

Case-II: For any thermodynamical process from one to another ~~systems~~ of a closed systems.

$$Q_{12} - W_{12} = U_2 - U_1$$

$$Q_{12} = W_{12} + U_2 - U_1.$$

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